

Week 1

Date: August 2026

Goal

I wanted to understand how launch angle and initial velocity affect a rocket's flight path.

What I Worked On

- Created the GitHub repository.
- Designed the project structure.
- Researched projectile motion equations.
- Identified the variables needed for the simulation.

What I Learned

I learned that a rocket's trajectory can be estimated using basic physics equations when air resistance is ignored.

Challenges

At first, I found it difficult to understand how horizontal and vertical velocity components work together to create a flight path.

Next Steps

- Create the first working version of the simulator.
- Calculate maximum height, flight time, and range.

Week 2

Date: August 2026

What I Worked On

- Developed the first Python version of the calculator.
- Implemented calculations for maximum height.
- Implemented calculations for flight time.
- Implemented calculations for horizontal range.

What I Learned

I learned how a small change in launch angle can significantly affect the trajectory of a rocket.

Results

The program successfully calculated rocket performance for different launch conditions.

Challenges

Testing different scenarios and making sure the calculations matched expected physics results.

Next Steps

- Add graphs and visualizations.
- Improve the user interface.

Week 3

Date: August 2026

What I Worked On

- Added trajectory visualization using Python.
- Created the first flight-path graph.
- Improved the presentation of the simulation results.

What I Learned

Visualizing the trajectory made the physics much easier to understand and helped me see the relationship between angle, range, and altitude.

Results

The simulator can now display rocket trajectories in addition to numerical results.

Key Lessons Learned

- Physics can be modeled using programming.
- Mathematics is a powerful tool for understanding real-world engineering problems.
- Small changes in input values can produce very different outcomes.
- Engineering combines science, mathematics, and technology to solve practical challenges.

Week 4

Date: September 2026

What I Worked On

- Reviewed the simulator calculations and tested different launch scenarios.
- Compared how changes in launch angle affect altitude, flight time, and range.
- Improved the project documentation and README file.
- Added visual materials and project explanations for GitHub.

What I Learned

I learned that engineering projects are not only about writing code. Testing, documenting, and presenting results are just as important as developing the calculations themselves.

I also realized that the same mathematical model can produce very different outcomes depending on the initial conditions. Even small adjustments in launch angle can significantly change a rocket's performance.

Results

Completed the first full version of the Rocket Trajectory Simulator.

Added project visuals and documentation.

Organized the project into a format suitable for a public engineering portfolio.

Challenges

One challenge was making the project easy to understand for people without a physics background. I wanted the calculations, graphs, and explanations to be both technically correct and easy to follow.

Next Steps

- Explore how air resistance affects rocket trajectories.
- Compare launch performance on Earth, the Moon, and Mars.
- Investigate more advanced aerospace simulation concepts.

Key Lessons Learned from the Entire Project

- Physics concepts can be transformed into working engineering tools through programming.
- Engineering is not only about calculations but also about testing, communication, and continuous improvement.
- Data visualization makes complex concepts much easier to understand.
- Small changes in design inputs can create large differences in performance.
- Curiosity and experimentation are essential parts of learning engineering.

Final Reflection

This project was my first opportunity to combine programming, mathematics, and Aerospace Engineering concepts in a practical way. It helped me better understand projectile motion while improving both my Python and problem-solving skills.

The most rewarding part was seeing a physics concept from a textbook turn into a working simulation that I built myself.

Small changes can completely change a mission.